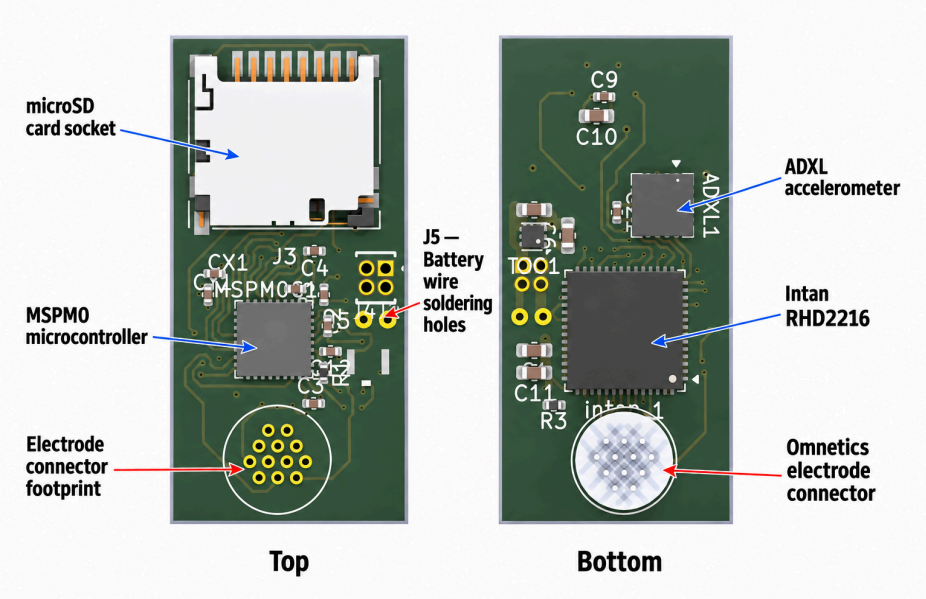
Rat Neural Data Logger and Debugger

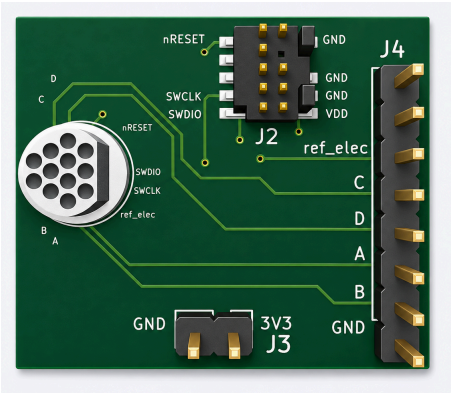
SYSTEM OVERVIEW

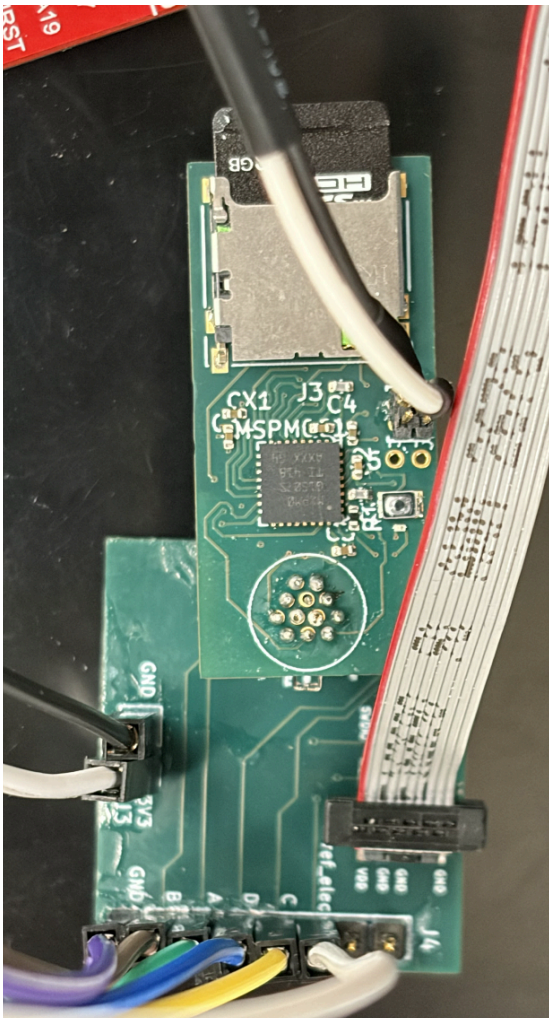
Designed and implemented a four-channel neural data logger using the TI MSPM0G1507 and Intan RHD2216, with 20 kHz sampling per channel, SPI DMA acquisition, triple-buffered microSD storage, and concurrent data acquisition and recording.

Validated 21 h 38 min of continuous diagnostic recording, verifying over 1.55 billion neural frames with zero detected sequence discontinuities, buffer overruns, or skipped DMA acquisitions.

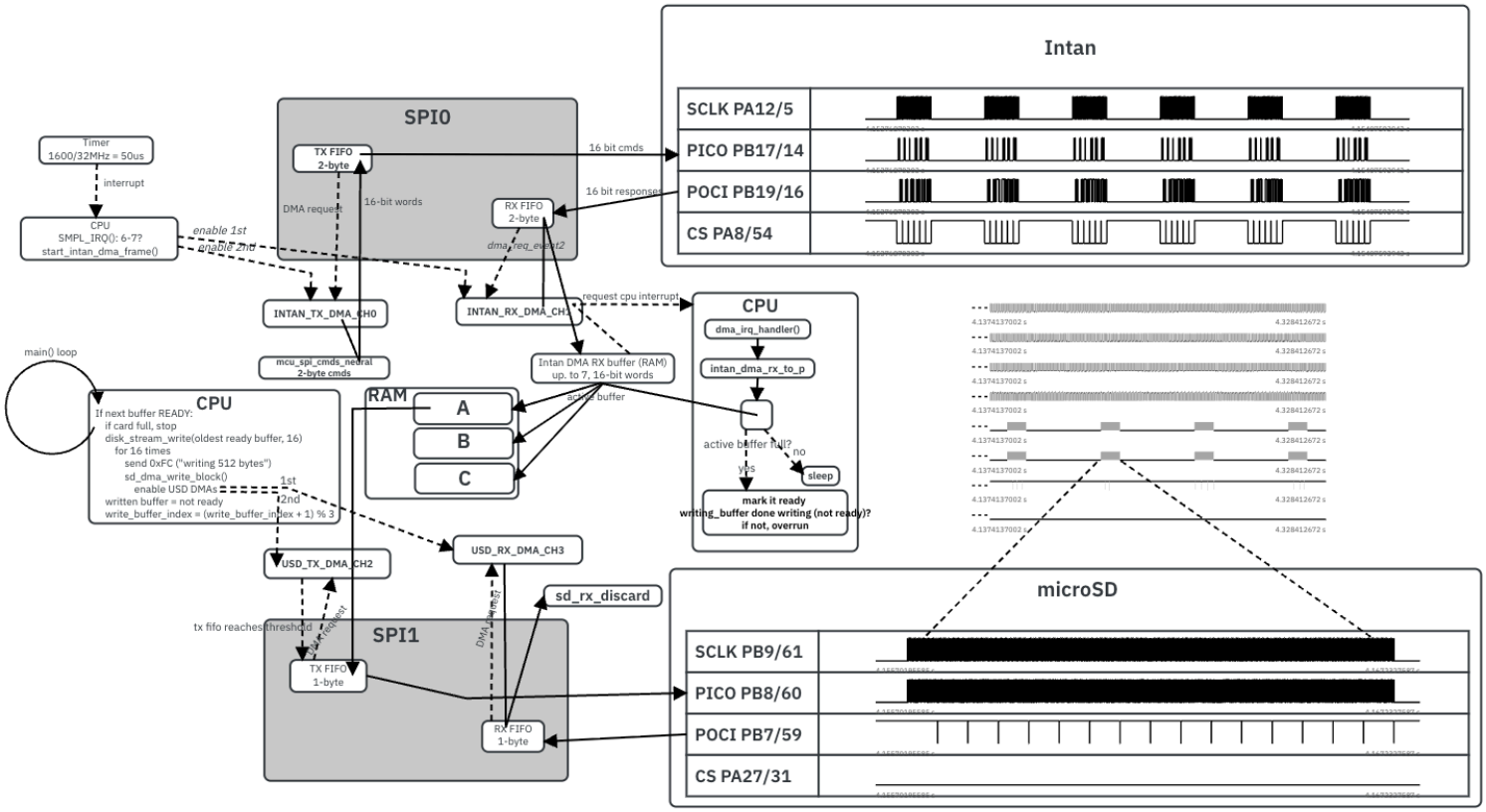
Measured assembled-logger power consumption of 61.495 mW (0.061495 W) at 3.3 V using a Kexin 16 GB microSD card. Input-referred noise and calibrated neural signal fidelity remain under validation.





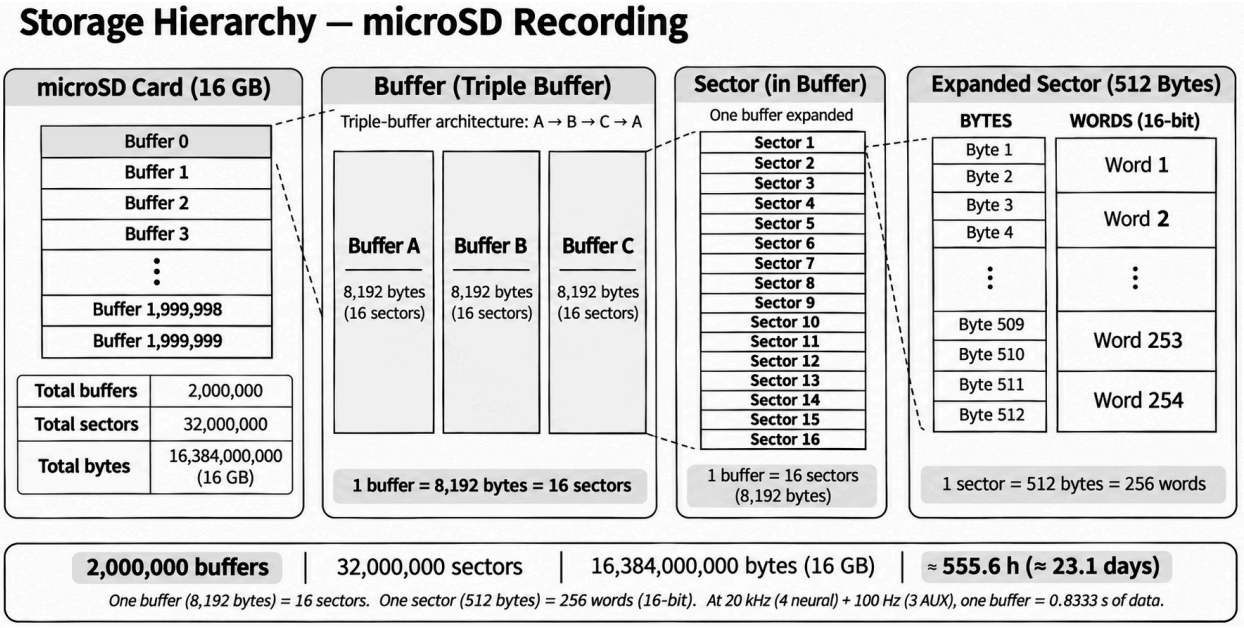


Logger Firmware



- **Acquisition:** A 20 kHz hardware timer triggers Intan SPI acquisition every 50 µs. SPI0 TX/RX DMA transfers neural data without CPU-driven SPI polling.
- **Neural channels:** Four channels sampled at 20 kHz each; three auxiliary channels sampled at 100 Hz each.
- **Buffering:** The CPU transfers completed samples into three 8,192-byte RAM buffers (24,576 bytes total), switching buffers automatically when full.
- **Storage:** SPI1 TX/RX DMA writes completed buffers directly to microSD using continuous CMD25 multi-block transfers.
- **Concurrent operation:** Neural acquisition continues while the SD controller writes previously completed buffers.
- **Data integrity:** FIFO ordering checks, buffer-overrun detection, DMA skip counters, timer-gap monitoring, and CRC32-protected checkpoints every ~10 seconds.
- **Diagnostic mode:** Optional 16-bit acquisition sequence counter stored per neural frame enables offline verification of frame continuity, with 25% additional neural data throughput.
- **Failure handling:** Recording stops before overwriting an unwritten RAM buffer or after a detected SD write failure.

MicroSD Storage



- **Triple buffering:** Three 8,192-byte RAM buffers (24,576 bytes total) operate in chronological A → B → C order.
- **Sector organization:** Each buffer contains 16 sectors of 512 bytes, transferred using SPI DMA.
- **Continuous recording:** CMD25 multi-block writes allow acquisition and SD storage to operate concurrently.
- **Buffer protection:** A buffer remains unavailable for reuse until its SD write completes.
- **Diagnostic recording:** Four 16-bit neural samples plus one 16-bit sequence counter per frame, with three auxiliary samples every 200 acquisition ticks.
- **Measured capacity:** A Kexin 16 GB microSD recorded approximately 21.64 hours before reaching the end of its storage capacity.
- **Data verification:** Over 1.5 billion neural frames were checked, with zero detected sequence discontinuities.

Data

HARDWARE SPECIFICATIONS

Microcontroller	TI MSPM0G1507
Neural amplifier	Intan RHD2216
Neural recording channels	4
Neural sampling rate	20,000 samples/s/channel
Auxiliary channels	3
Auxiliary sampling rate	100 samples/s/channel
ADC resolution	16-bit
Voltage resolution	0.195 µV/count
Storage	microSD (raw SPI writes)
Buffer architecture	Triple buffer, 8,192 bytes each
Production recording format	4 neural samples/frame + 3 AUX samples/200 frames
Diagnostic recording format	4 neural samples + 16-bit sequence/frame
Diagnostic data overhead	25% additional neural payload
Data transfer	SPI + DMA
Supply voltage	3.3 V nominal

POWER CONSUMPTION — ASSEMBLED LOGGER PCB

Supply voltage	3.3 V nominal
Current measurement method	0.15 Ω shunt resistor, 100× amplifier gain
Analog sampling rate	1,562,500 samples/s
MicroSD card	Kexin 16 GB
Capture duration	64.557 s
Average current	18.635 mA (0.018635 A)
Steady-state current (5–55 s)	18.667 mA (0.018667 A)
Steady-state power (5–55 s)	61.601 mW (0.061601 W)
Average power	61.495 mW (0.061495 W)
Peak current	74.000 mA (0.074000 A)
Total charge	1.203007 C
Total energy	3.969922 J

CLOCK AND SAMPLING PERFORMANCE

MCU clock frequency	32 MHz
Neural sampling rate	20,000 samples/s/channel
Nominal sampling interval	50 µs (0.000050 s)
Maximum measured sampling interval	50.375 µs (0.000050375 s)
Measured Intan SPI transfer (neural only)	12.95 µs (0.00001295 s)
Measured Intan SPI transfer (with AUX)	19.46 µs (0.00001946 s)
Measured SD buffer write time (development setup)	10.20 ms (0.01020 s)
Maximum SD write time (assembled logger, Kexin)	18.65 ms (0.01865 s)
Sampling timing gaps (21.64 h diagnostic test)	0
Skipped Intan DMA acquisitions (21.64 h test)	0

ANALOG FRONT-END SPECIFICATIONS

Neural amplifier	Intan RHD2216
Active neural channels	4
ADC resolution	16-bit
Voltage resolution	0.195 µV/count (0.000000195 V/count)
Analog upper bandwidth	Approximately 5 kHz (firmware setting)

MECHANICAL SPECIFICATIONS

PCB length	32.96 mm (0.03296 m)
PCB width	14.75 mm (0.01475 m)
PCB layer count	4
Neural connector	Omnetics A22016-001
microSD socket	DM3D-SF
Recording button	GSNANOT000102
Battery model	PKCELL 503035
Battery chemistry	Lithium-polymer
Battery capacity	500 mAh (0.5 Ah)
Battery nominal voltage	3.7 V
Battery protection	Integrated protection circuit (PCM)
Battery dimensions	35 × 30 × 5 mm

DATA INTEGRITY AND ERROR HANDLING

Buffer architecture	3 × 8,192 bytes (24,576 bytes total)
Buffer write order	Chronological FIFO (A → B → C)
Buffer overrun protection	Stops acquisition before overwriting unwritten data
Checkpoint interval	Approximately 10 seconds
Checkpoint integrity	CRC32
Buffer sequence verification	Sequential block-number checks before SD writes
Frame sequence verification	Optional 16-bit acquisition counter per neural frame
Sequence counter wraparound	Every 65,536 acquisition ticks
Offline frame verification	Detects missing, duplicated, and out-of-order sequence numbers
SD write error handling	Stops recording on detected write failure
SD busy timeout	1,000,000 polling iterations
Longest diagnostic recording	21 h 38 min 37.64 s
Missing frames detected	0
Frame sequence discontinuities	0
Buffer overruns	0
Skipped DMA acquisitions	0
Reported sampling gaps	0
Valid checkpoints	7,787
DMA started minus completed	1 at final checkpoint
Recording termination	Near card capacity; exact cause unconfirmed

Recording Performance: Kexin 16GB — Diagnostic Mode

Recording duration	21 h 38 min 37.64 s
Neural channels	4
Sampling rate	20,000 samples/s/channel
Sequence counter	16-bit per neural frame
Frames verified through last checkpoint	1,558,207,425
Additional verified tail frames	145,382
Last verified acquisition tick	1,558,352,807
Missing frames detected	0
Sequence errors	0
Buffer overruns	0
Skipped DMA acquisitions	0

Reported sampling gaps	0
Valid checkpoints	7,787
SD writes >51.2 ms	1
Maximum SD write duration	59.40 ms
Maximum checkpoint gap	10.013 s
DMA started minus completed	1 at final checkpoint
Remaining erased sectors	5 (2,560 bytes)
Recording stop cause	Near card capacity; unconfirmed

Recording Performance: SanDisk Ultra 16GB — Production Format	
Recording duration	2 h 20 min 55.7 s
Neural channels	4
Sampling rate	20,000 samples/s/channel
Buffer overruns	0
Skipped DMA acquisitions	0
Reported sampling gaps	0
Completed DMA acquisitions	169,113,282
Valid checkpoints	845
SD writes >51.2 ms	31
Maximum SD write duration	74.85 ms
SD busy events >10 ms	381

Recording Performance: Kingston 16GB	
Recording duration	11 h 11 min 50.3 s
Neural channels	4
Sampling rate	20,000 samples/s/channel
Buffer overruns	0
Skipped DMA acquisitions	0
Reported sampling gaps	0
Completed DMA acquisitions	806,205,021
Valid checkpoints	4,027
SD writes >51.2 ms	0
Maximum SD write duration	30.30 ms
SD busy events >10 ms	4

Recording Performance: Kexin 16GB — Assembled Logger	
Recording duration	68.4845 s
Neural channels	4
Sampling rate	20,000 samples/s/channel
Neural frames in packed data	1,369,600
Buffer overruns	0
Skipped DMA acquisitions	0
Reported sampling gaps	0
Valid checkpoints	7
SD writes >51.2 ms	0
Maximum SD write duration	18.65 ms
SD busy events >10 ms	0
DMA started	1,369,690
DMA completed	1,369,690

Repeated Recording Performance: Kexin 16GB — Assembled Logger	
Recording A duration	19.9696 s

Recording B duration	27.9737 s
Recording C duration	32.38415 s
Recording D duration	26.50355 s
Recording A neural frames	399,200
Recording B neural frames	559,400
Recording C neural frames	647,600
Recording D neural frames	530,000
Buffer overruns across all recordings	0
Skipped DMA acquisitions across all recordings	0
Reported sampling gaps across all recordings	0
Maximum SD write duration across all recordings	22.70 ms

NEXT VALIDATION MEASUREMENTS

Input-referred noise and noise spectrum under verified input conditions

Calibrated microvolt-level signal recovery and SNR

Battery runtime using PKCELL 503035 500 mAh (0.5 Ah) battery

Long-term clock accuracy and sampling jitter

Repeat frame-sequence verification across multiple cards and long-duration recordings

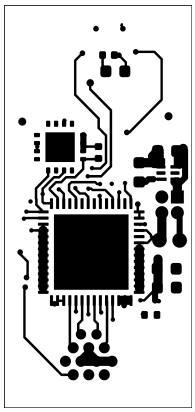
Final mass, encapsulation, and connector strain-relief validation

Preliminary noise results excluded pending verified electrode connections and repeat testing

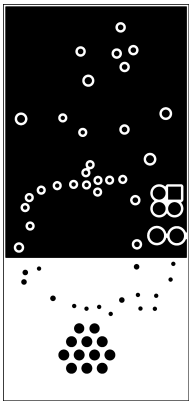
Design files

Four KiCad files for logger, four KiCad files for debugger, Logger and debugger 3d model files and footprints:
https://drive.google.com/file/d/1VDEnnRkDs59kNcbnMssBrENyT8biDzK2/view?usp=sharing
https://drive.google.com/file/d/1F0bLwUQ0UwHwFkUPgPWslSSLJwXrTadQ/view?usp=sharing
https://drive.google.com/drive/folders/1Eqr3tGwiLbXq0Q-8Us4gilH18KxEu-P2?usp=drive_link
Contact dlk96@case.edu for firmware

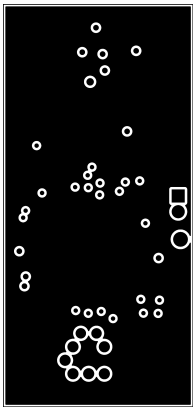
FCu



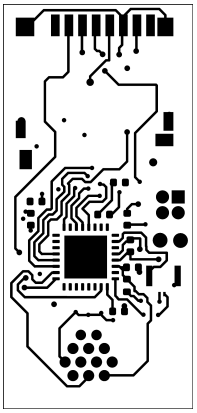
3v3



GND



BCu



Sch

